

**A TECHNICAL REPORT ON
STUDENT INDUSTRIAL WORK EXPERIENCE SCHEME (SIWES)**

**CARRIED OUT AT
MODIBBO ADAMA UNIVERSITY,
YOLA, ADAMAWA STATE**

**FROM: AUGUST, 2025
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BY

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**A SIWES REPORT SUBMITTED TO THE DEPARTMENT OF FOOD
SCIENCE TECHNOLOGY, SCHOOL OF APPLIED SCIENCE,
FEDERAL POLYTECHNIC, MUBI, ADAMAWA STATE, NIGERIA**

**IN PARTIAL FUFILMENT OF THE REQUIREMENTS FOR THE
AWARD OF NATIONAL DIPLOMA (ND) IN FOOD SCIENCE
TECHNOLOGY,**

FEBRUARY, 2026

APPROVAL PAGE

This is to certify that Liuramasa Orpha with a registration number SAS/FST/ND/24/004 has undergone his student industrial training at **Modibbo Adama University, Yola, Adamawa State**. In partial fulfilment for the award of national diploma (ND) in Food Science Technology Federal Polytechnic Mubi as certified by the undersign officer.

.....

MR. USMAN U. WAFARI

DEPERTMENTAL SIWES COORDINATOR

.....

DATE

DEDICATOIN

This report is dedicated to God almighty for this wisdom and strength and my lovely parents for their love and care throughout the period of my SIWES training.

ACKNOWLEDGEMENT

I thank God Almighty for making me to undergo students industrial work experience scheme (SIWES) successfully.

My gratitude goes to my lovely parents for their prayers, financial and moral support during my attachment.

I also appreciate the kind gesture of my brothers and sisters and those who supports me in prayers and contribution during my industrial attachment.

I whole heartedly thank my Head of Department, SIWES coordinator and all lecturers of Food Science Technology Department for their effort to ensure my success as their students.

I am greatly indebted to my co-SIWES students to mention, may God strengthen our relationship together and grant us academic excellence.

I sincerely thank you all for your contribution and support.

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CHAPTER ONE

1.0 INTRODUCTION

The concept behind industrial training schemes is to acquire practical, knowledge in addition to what has been learn institution. The four months mandatory supervised industrial attachment is program instituted by the federal government to help student for their different field of studies.

1.1 Background of SIWES

It aims at exposing student to the practical and the actualization of work situation which they may meet after graduation and the learning is meanly expected to produce graduate qualification to meet man power of employments of (ITF), there was growing concern amount the industrialist that graduate of higher learning do not passes adequate background student for the opinion that theoretical education going higher institution was responsive enough to meet the need of employers of the labour as a requirement of national board for technical education (NBTE) polytechnic training before obtaining the National Diploma (ND). Experience scheme (SIWES), is a National programme introduced by Government in 1974 for student in tertiary institution.

1.2 Brief History of SIWES

SIWES was established by **ITF** in 1973 to solve the problem of lack of adequate practical skills preparatory for employment in industries by Nigerian graduates of tertiary institutions.

The program “Student industrial work experience scheme” is a pure skill-acquisition program structured for the tertiary institutions as a complementary and enhancement program to the theoretical education, laboratory and workshop practices engaged in by student in different higher institutions. The SIWES was initiated to improve the student’s technical abilities (performance/knowledge) and expose them to industrial culture thereby preparing them to be acquainted with the roles to play towards the technological advancement of the nation.

It is there for a practical aspect of the academic works, which the students may not be opportune to carry out throughout their stay in the higher institutions. Based on this fact, the Federal Government decided to establish a body (regulatory) which engages in the training of the

technical manpower. The body is named industrial Training Fund (ITF). The government also undertook to make up for the deficiencies by structuring and established Students Industrial Training (SIT) as it was then called but in 1973 the SIWES was formed which is the subsidiary.

1.3 Objectives of SIWES

- One of the objectives of SIWES is to create an avenue for students to acquire industrial skills to complement their theoretical knowledge and improve their experiences in their course of study.
- The program prepares the students for industrial working conditions prior to their graduation.
- To enable the students to learn personal relationship with employers and co-employees on graduation.
- It exposes the student to working methods and techniques in handling equipment and machinery.
- It develops the student in make critical and realistic approaches to solving problem.
- It serves as a way of improving the abilities of the students and to contribute to the growth of the nation.
- It gives the students the opportunity to practice what they have been taught in various disciplines.
- The SIWES program strengthens the employer's involvement in preparing the students for employment.

CHAPTER TWO

2.1 Brief History of Modibbo Adama University Yola

The University is located at Sangere, a village in Girei Local Government Area of Adamawa State, some 10km from Jimeta. In 1980 the Federal Government in response to the nationally acclaimed need for skilled, innovative, and technologically oriented high-level manpower required to develop and nurture a technology-based economy and society established seven new Universities of Technology. Federal University of Technology, Yola, along with those of Abeokuta, Akure, Bauchi, Makurdi, Minna, and Owerri, came into being in 1981. This University admitted its first batch of students (214 numbers) in the 1982/83 academic session. The Current population of regular undergraduates is 14600.

The University barely took off when the succeeding Military Government abolished four of the seven new Universities of Technology and merged them with older Universities in October 1984. Federal University of Technology, Yola was merged with the University of Maiduguri and became known as Modibbo Adama College of the University of Maiduguri. Four years later, on 1st January 1988, the University was granted its former name and full autonomy as the Federal University of Technology, Yola. A substantive Vice-Chancellor was therefore appointed to run the affairs of the University.

The University from its inception has run the School system of Administration with two Schools, the School of Science and Technology Education and the School of Management Sciences. Presently, the University has six Schools: the School of Agriculture and Agricultural Technology, Engineering and Engineering Technology, Environmental Sciences, Management and Information Technology, Pure and Applied Sciences, and Technology and Science Education.

The then President, Dr. Goodluck Ebele Jonathan, GCFR, approved the change of name of the University from the Federal University of Technology, Yola (FUT, Yola) To Modibbo Adama University of Technology, Yola (MAUTECH, Yola), supposedly named after Modibbo Adama

Ibn Hassan (1809-1847), a great scholar, an erudite educationist, an outstanding leader and the founder of the Fombina Kingdom (The Present Adamawa Emirate).

As the first ruler and founder of the Emirate, Modibbo Adama Ibn Hassan was one of the disciples and flag bearers of Sheikh Usman Ibn Fodio of the Sokoto Caliphate. The new name took effect from 1st October 2011. Subsequently, President Muhammadu Buhari assented to the Modibbo Adama University Yola establishment bill on the 6th of January 2021.

Vision

To be a leading center for advanced research that will spearhead ground-breaking novel scientific and technological discoveries and innovations for an all-encompassing national development.

Mission

To provide a multidisciplinary, multifaceted, and inclusive approach to advanced research for the attainment of the highest research potentials in science and technology for a knowledge-driven socio-economic and capital development of the nation.

2.2 Organizational Chart

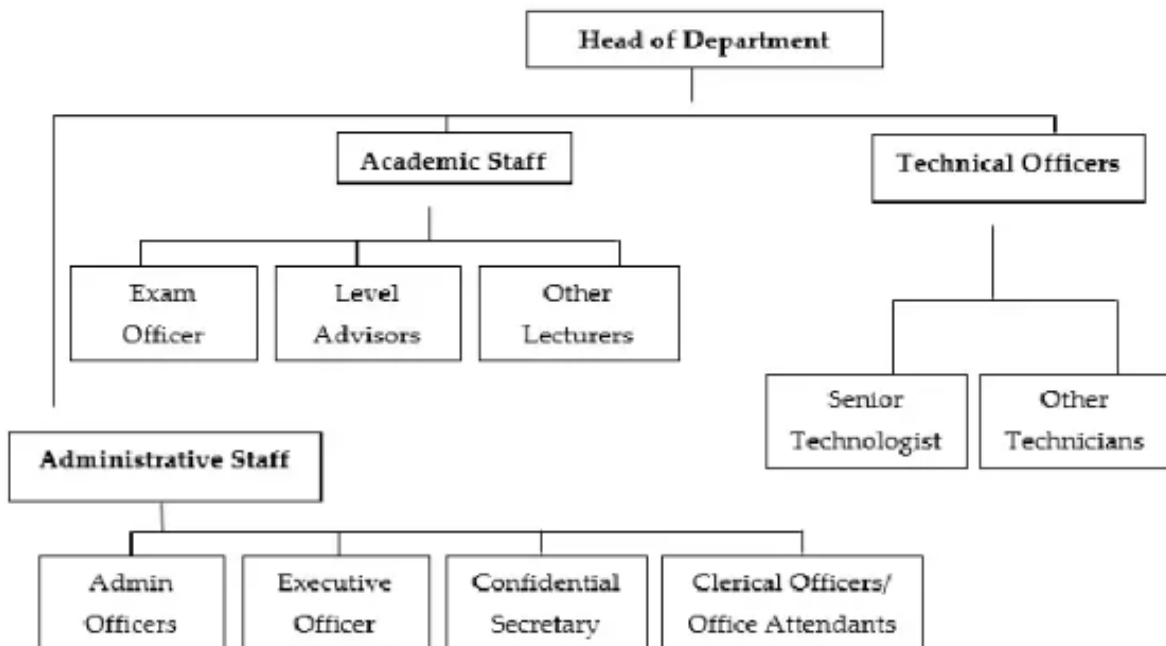


Figure 2.1: Organizational chart

CHAPTER THREE

EXPERIENCED GAINED DURING THE INDUSTRIAL TRAINING

3.1 Functional Properties

Functional properties refer to the physicochemical characteristics of food materials that determine their behavior during processing, storage, preparation, and consumption. In food science, these properties are especially important in understanding how food ingredients interact with water, oil, heat, and other components in food systems. During the Students Industrial Work Experience Scheme (SIWES), practical exposure was gained on the evaluation and application of functional properties of various food materials, particularly flours and plant-based food products. These properties play a crucial role in food formulation, product development, and quality control within the food industry.

3.1.1 Importance of Functional Properties in Food Processing

Functional properties are vital in the food industry because they:

- i. Influence food texture, consistency, and appearance
- ii. Determine the processing behavior of food materials
- iii. Aid in selecting suitable raw materials for specific food products
- iv. Improve product quality and consumer acceptability
- v. Support innovation and development of new food products

3.1.2 Objectives of Studying Functional Properties

The study of functional properties during SIWES was aimed at:

- i. Understanding the behavior of food materials during processing and preparation
- ii. Determining the suitability of food ingredients for specific food products
- iii. Enhancing product quality, texture, stability, and consumer acceptability
- iv. Applying theoretical knowledge of food chemistry to practical food processing operations
- v. Assisting in quality control and product standardization

3.1.3 Water Absorption Capacity (WAC)

Water absorption capacity refers to the ability of a food material to absorb and retain water. It is an important property in products such as bread, dough, soups, and sauces. High water absorption capacity improves dough handling and enhances product yield. During SIWES, water absorption capacity was determined by mixing a known quantity of flour with water, centrifuging the mixture, and measuring the amount of water retained by the sample.

- A known weight of the flour sample was accurately weighed into a clean, dry centrifuge tube.
- A measured volume of distilled water was added to the flour sample in the centrifuge tube.
- The mixture was stirred or shaken thoroughly to ensure complete wetting and uniform mixing of the sample.
- The mixture was allowed to stand for a specified time to enable water absorption.
- The tube was centrifuged at a suitable speed for a specified duration to separate the absorbed water from the unabsorbed water.
- The supernatant was carefully decanted without disturbing the sediment.
- The weight of the sediment was measured, and the amount of water retained by the sample was calculated.

3.1.4 Oil Absorption Capacity (OAC)

Oil absorption capacity is the ability of food materials to bind and retain oil. This property is important in flavor retention and mouthfeel, especially in products like sausages, snacks, and baked goods. Materials with high oil absorption capacity are desirable for improving palatability and flavor stability in food products.

- A known weight of the flour sample was weighed into a clean, dry centrifuge tube.
- A measured volume of edible oil was added to the sample in the centrifuge tube.

- The mixture was stirred or shaken thoroughly to ensure uniform contact between the flour and the oil.
- The mixture was allowed to stand for a specified time to enable oil absorption.
- The tube was centrifuged at an appropriate speed for a specified duration to separate absorbed oil from free oil.
- The supernatant oil was carefully decanted without disturbing the sediment.
- The weight of the oil absorbed by the sample was determined.
- Oil absorption capacity was calculated and expressed as the amount of oil absorbed per gram of flour sample.

3.1.5 Solubility Determination

Solubility determination refers to the ability of a food material to dissolve or disperse in a solvent, usually water, under specific conditions. It is an important functional property in food science because it influences product texture, appearance, processing behavior, and consumer acceptability. Solubility is especially important in products such as beverages, soups, sauces, and instant food formulations. During SIWES, solubility determination was carried out to evaluate how well food samples, particularly flours and powders, dissolve in water.

- A known weight of the food sample was accurately weighed using an analytical balance.
- A measured volume of distilled water was added to the sample in a clean beaker or test tube.
- The mixture was stirred or shaken thoroughly to ensure proper dispersion of the sample in the water.
- The suspension was allowed to stand for a specified time or subjected to centrifugation to separate soluble components from insoluble residues.
- The clear supernatant was carefully decanted or pipetted into a clean, pre weighed container.

- The supernatant was dried in an oven at a specified temperature until a constant weight was obtained.
- The dried residue was weighed, and the weight of soluble solids was determined.
- Solubility was calculated as the percentage of soluble solids relative to the original weight of the food sample.

3.1.6 Bulk Density

Bulk density refers to the mass of a food material per unit volume. It affects packaging, storage, transportation, and product formulation. Low bulk density is desirable in weaning foods, while higher bulk density may be preferred in bakery products.

- A clean and dry graduated measuring cylinder was selected and its volume was noted.
- The empty measuring cylinder was weighed and the weight was recorded.
- The flour sample was gently poured into the measuring cylinder without tapping or compacting.
- The cylinder was filled to a known volume and the surface of the sample was leveled off.
- The measuring cylinder containing the flour sample was weighed and the weight was recorded.
- The weight of the flour sample was obtained by subtracting the weight of the empty cylinder from the weight of the filled cylinder.
- Bulk density was calculated by dividing the weight of the flour sample by the volume it occupied and expressed in grams per milliliter.

3.1.7 pH Determination using a pH Meter

pH determination refers to the measurement of the acidity or alkalinity of a food sample and is an important quality parameter in food science. The pH of food materials influences flavor, shelf life, microbial stability, and processing behavior. It also affects enzyme activity, protein functionality, and overall product quality. During SIWES, pH determination was carried out using a digital pH meter to assess the chemical nature of various food samples.

- The pH meter was first switched on and allowed to warm up for proper functioning.
- The instrument was calibrated using standard buffer solutions, usually pH 7.0 and pH 4.0, to ensure accuracy of readings.
- A known quantity of the food sample was weighed and mixed with distilled water to form a uniform suspension or solution.
- The electrode of the pH meter was rinsed with distilled water and gently blotted dry to remove excess water.
- The electrode was immersed into the prepared food sample, ensuring that it was fully covered by the solution.
- The pH reading was allowed to stabilize on the display screen before being recorded.
- After recording the reading, the electrode was removed and rinsed with distilled water to prevent contamination.
- The pH meter was switched off and the electrode was properly stored in the recommended storage solution.

3.2 Proximate Analysis

Proximate analysis is a fundamental analytical procedure in food science used to determine the basic nutritional composition of food materials. It involves the quantitative determination of moisture content, ash content, crude protein, crude fat, crude fibre, and carbohydrate by difference. During the SIWES programme, proximate analysis was carried out to assess the nutritional quality, shelf stability, and suitability of food samples for consumption and processing.

3.2.1 Importance of Proximate Analysis

Proximate analysis plays a vital role in food science, nutrition, and quality assurance. One of its major importance is in **evaluating the nutritional adequacy of food materials**, as it provides essential information on the levels of major nutrients required for human growth, energy, and maintenance. This information is useful for diet planning, nutritional assessment, and public health studies.

Additionally, proximate analysis is important for **quality control and standardization** in food processing industries. By determining moisture, fat, protein, and ash contents, producers can maintain consistent product quality, detect adulteration, and ensure compliance with national and international food safety standards.

3.2.2 Objectives of determining Proximate Analysis

The objectives of carrying out proximate analysis include determining the nutritional value of food materials, ensuring compliance with food quality standards, supporting product formulation, and providing essential data for labelling and quality control purposes.

3.2.1 Carbohydrate Content

- The flour sample was first properly mixed to obtain a uniform and representative sample for analysis. A known weight of the flour was weighed using an analytical balance and transferred into a clean test tube.

- Distilled water was added to the flour sample and the mixture was thoroughly shaken to extract soluble carbohydrates. The mixture was then allowed to stand or filtered to obtain a clear extract.
- Using a pipette, a measured volume of the flour extract was transferred into a clean test tube.
- A measured volume of phenol solution was carefully added to the test tube using a pipette, and the mixture was gently shaken to ensure proper mixing.
- Concentrated sulphuric acid was then carefully added down the side of the test tube using a pipette, allowing it to mix with the solution without splashing. The test tube was left undisturbed for a short period to allow colour development.
- The mixture was gently shaken and allowed to cool, during which a yellow to orange colour developed, indicating the presence of carbohydrates.
- The intensity of the colour formed was measured using a spectrophotometer at the appropriate wavelength, and the absorbance reading was recorded.
- The carbohydrate content of the flour sample was determined by comparing the absorbance value with that of standard glucose solutions prepared under the same conditions.
- The final carbohydrate concentration was calculated and recorded as the carbohydrate content of the flour sample.

3.3 PROCESSING OF BAKERY PRODUCTS

Food processing involves the transformation of raw materials into edible and acceptable food products through various physical, chemical, and biological methods. In food science, bakery processing is an important area that involves the use of flour and other ingredients to produce products such as biscuits, cookies, and bread. During the SIWES programme, practical exposure was gained in the processing of these bakery products, with emphasis on ingredient formulation, mixing, baking, and quality evaluation.

3.3.1 Processing of Biscuits

Biscuit processing involves the production of dry, crisp baked products with low moisture content and long shelf life. The major ingredients used include wheat flour, sugar, fat, baking powder, salt, and water.

Steps in Biscuit Processing

- Raw materials such as flour, sugar, fat, baking powder, and salt were weighed accurately.
- The dry ingredients were mixed thoroughly in a mixing bowl.
- Fat was added and mixed until a uniform mixture was obtained.
- Water was added gradually to form a smooth dough.
- The dough was rolled out to the required thickness.
- Dough pieces were cut into desired shapes using cutters.
- The shaped dough was placed on baking trays.
- The biscuits were baked in an oven at a controlled temperature until golden brown.
- The baked biscuits were cooled and packaged.

3.3.2 Processing of Cookies

Cookies are sweet baked products that are generally softer and richer in fat and sugar compared to biscuits.

Steps in Cookies Processing

- Fat and sugar were weighed and creamed together until light and fluffy.
- Eggs and flavoring agents were added and mixed thoroughly.
- Flour and leavening agents were gradually added to form a soft dough.
- The dough was portioned and shaped as required.
- The shaped dough was arranged on baking trays.
- The cookies were baked in an oven at the appropriate temperature.
- The baked cookies were removed from the oven and allowed to cool.
- The cooled cookies were packaged.

3.3.3 Processing of Bread

Bread processing involves fermentation and baking to produce a soft, porous food product. The main ingredients include wheat flour, yeast, water, sugar, salt, and fat.

Steps in Bread Processing

- Ingredients such as flour, yeast, sugar, salt, fat, and water were accurately weighed.
- All ingredients were mixed to form dough.
- The dough was kneaded to develop gluten structure.
- The dough was allowed to ferment for a specified time.
- The fermented dough was punched down to remove excess gas.
- The dough was molded and shaped into loaves.
- The shaped dough was proofed until it increased in size.
- The proofed dough was baked in an oven at the required temperature.
- The baked bread was cooled, sliced, and packaged.

CHAPTER FOUR

SUMMARY, CONCLUSION AND RECOMMENDATIONS

4.1 Summary

The relevance of the SIWES program cannot be over emphasized considering the fact that it has significantly reduced the gap between my theoretical and practical knowledge about computer hardware and software, installations, maintenance and networking. The processes of communication which include data and telecommunication the use of switch in the networking and what networking is all about.

The program is indeed a commendable one in that it affords students ample opportunities of being exposed to good working relationship with colleagues and the field experience with customers. This little exposure has widened my knowledge about my course of study, not only that it has automatically changed my views about life in general. The firm at large has taught me how to be independent of my own how to be conscious of my health and safety at its peak relating to the environment where I carried out my SIWES program. It was indeed a highly rewarding experience to be with Modibbo Adama University, Yola, Adamawa State.

4.2 Conclusion

In conclusion, I thank ITF in general for their effort towards the Student Industrial Training Scheme. The contribution that the industrial training offered to student will not be over emphasized. It has exposed me seriously to a certain depth and length of practical capability on Computer Knowledge and practical Know how.

It has also acquainted me with the working condition, which I am expected to encounter in the near future. I will say that SIWES has a greater advantage on me, it has greatly exposed me to the practical application of all that I have been through in the school, SIWES is an experience that all student must pass through this is because it gives a full practical knowledge of what has been through in classroom.

Finally, I have a strong believe that this comprehensive based on the experience, I acquired during the industrial training scheme will convince every user training is not difficult.

4.3 Recommendations

Having gone through this training my recommendation are:

- i. The students should be well supervised, both from the ITF and institution
- ii. The institution supervisor should be from the students department so as to be able to dictate any fault.
- iii. The organization should make provision for some amount to stimulate the students monthly.
- iv. The students that are ready to learn should be well enlightened from the organization.

REFERENCES

- Adegboye, R. O. (2012). *Challenges of industrial training in Nigeria*. Lagos: Yaba College Press.
- Akinyemi, S. (2011). *Bridging the skills gap in Nigerian higher institutions: The SIWES approach*. Nigerian Journal of Educational Research, 14(3), 22–29.
- Industrial Training Fund (ITF). (2010). *SIWES Policy Document*. Jos: ITF Publications.
- Oladele, J. O., & Akintayo, M. O. (2013). *Effectiveness of the Students' Industrial Work Experience Scheme (SIWES) in Nigerian tertiary institutions*. Journal of Technical Education and Training, 5(1), 35–42.